Ishrak Hayet

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SUMMARY

Working on Applied Machine Learning research and development. Experienced with data collection, cleaning, and curation for training, fine-tuning, and evaluating large-scale optimization problems. Currently working on fine-tuning RL/LLMs to generate code from specifications and implementation behavior. Have strong foundations of machine learning, deep learning, and statistical models, supervised and unsupervised learning methods, and hypothesis testing. Proficient in object-oriented programming in Python and Java. Extremely efficient in collaborating with generative AI models, agents with manual validation for solving everyday tasks 10x faster. No sponsorship is required. Have approved I-140. Adjustment of status I485 submitted.

EDUCATION

PhD in Computer Science — 08/2022 - 12/2026
North Carolina State University • Raleigh, NC

MSc in Computer Science — 08/2019 - 05/2022
University of Kansas • Lawrence, KS

BSc in Computer Science and Engineering — 01/2014 - 12/2017
IUT • Dhaka, Bangladesh

EXPERIENCE

Research Scientist Intern — 06/2026 - 08/2026
AI for Code Lab, IBM Research, Yorktown Heights

- Development of Software Engineering Agents via program analysis, knowledge graph, symbolic reasoning, and constrained generation..
- Evaluating software agents on patch validation, automated program repair using SWEBench, SWEBench-Verified, etc. benchmarks.

Applied Machine Learning PhD Researcher — 08/2022 - 12/2026
WISER Lab, North Carolina State University • github.com/ncsu-swat

- Built a partial code execution technique using feedback-directed mock generation and neural type inference with 80 % coverage.
- Developed a prompt engineering framework to generate and repair JUnit test oracles using LLMs, beating SoTA accuracy by 10 %

Machine Learning Privacy MS Researcher — 08/2019 - 07/2022
InfoSec Lab, University of Kansas • github.com/ihayet/Invernet

- Built a deep neural model to infer private natural language datasets from pre-trained and fine-tuned embeddings with 70 % accuracy
- Trained Word2Vec, GloVe, and BERT vectors and built LSTM, CNN, and Transformer models for datasets with 20K samples

PROJECTS

Neuro-symbolic code completion and type prediction with AI ***(ACM ISSTA '24)***

- Developed a neuro-symbolic debugging technique for partially executing incomplete code
- Built a dataset of incomplete code, mined inductive mocking rules from error semantics, and fine-tuned an LLM for type prediction

Ground truth generation for unit tests using LLM workflows ***(IEEE TSE '25)***

- Implemented a few-shot prompting framework and LLM workflow for generating and repairing unit test assertion statements
- Evaluated on GPT3.5, CodeLlama, Mistral, and other open-source models - achieved high BLEU, CodeBLEU, Rouge, and mutation scores

Machine learning model to detect language model privacy leaks ***(EMNLP Findings '22)***

- Devised a differential vector embeddings inference model to expose word-to-word co-occurrence properties of a fine-tuning dataset
- Used greedy-search decoding to reconstruct fairly faithful samples from extracted word-to-word co-occurrences with high precision/recall

SKILLS

Development	C/C++, Python, Java, Rust, JavaScript – Node.js – PostgreSQL, Mongo – AWS, GCP, WebAssembly, Docker
Testing	JUnit, PIT, Jasmine, AFL++ – Maven, Gradle, Jenkins, Selenium, GH Workflows - A/B Testing
Machine Learning	Numpy, Pandas, PyTorch, Keras, Hugging Face, GPT, Llama, Mistral, LoRA, RLHF, LangChain, LangGraph, MCP